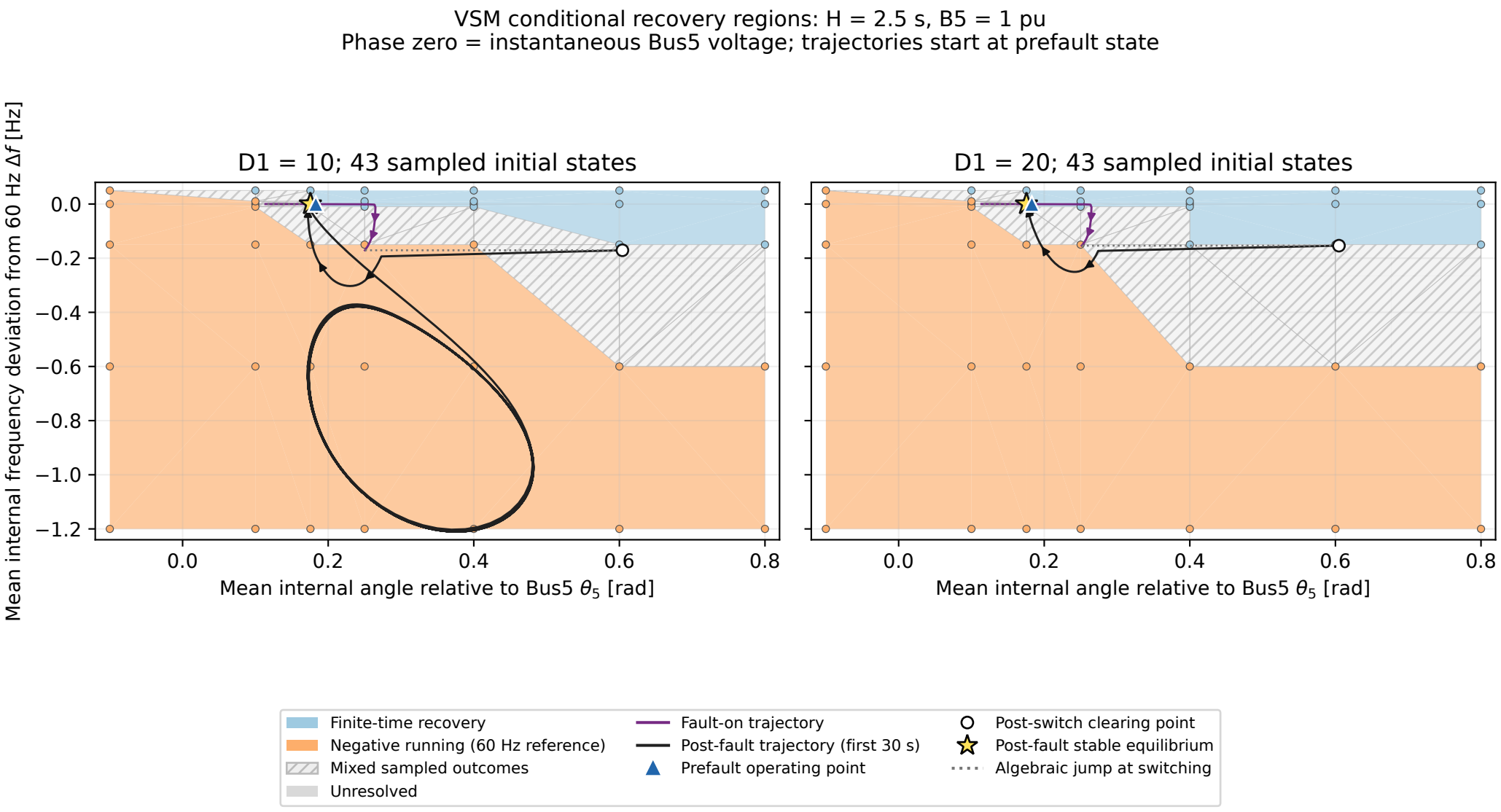
VSM conditional recovery regions: H = 2.5 s, B5 = 1 pu
Phase zero = instantaneous Bus5 voltage; trajectories start at prefault state
D1 = 10; 43 sampled initial states
D1 = 20; 43 sampled initial states
Mean internal frequency deviation from 60 Hz Δf [Hz]
Mean internal angle relative to Bus5 θ5 [rad]
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.2
0.4
0.6
0.8
Finite-time recovery
Negative running (60 Hz reference)
Mixed sampled outcomes
Unresolved
Fault-on trajectory
Post-fault trajectory (first 30 s)
Prefault operating point
Post-switch clearing point
Post-fault stable equilibrium
Algebraic jump at switching